

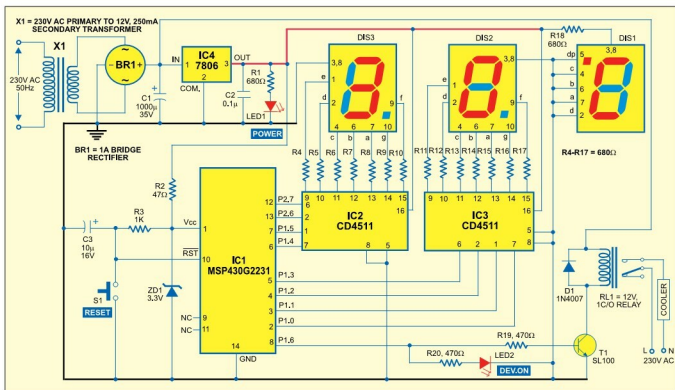


Fig. 2: Circuit diagram of the low-power temperature indicator and controller

Relay. As the I/O pin of the MCU cannot drive a relay, a transistor is used for this purpose. LED2 provides a visual indication of the relay status. Electrical appliances such as coolers or fans can be switched on/off through the relay. Glowing of LED2 will indicate the 'on' status of the electrical device connected across relay RL1.

Input switch. Here reset switch S1 is the only input switch. Press it momentarily whenever there is incorrect data display. Note that there is no control input switches for the temperature setting. The program code is fixed at 29°C. That is, if the temperature reaches 30 degrees, your electrical appliance (cooler, fan or AC) connected to this circuit will turn on automatically. If you want to change the temperature setting, you need to change the code as explained below in the software section.

Display. DIS2 and DIS3 display the temperature in digits from 00 to +99. Temperature is commonly measured either in degree centigrade or Fahrenheit. This project displays temperature in degree centigrade. DIS1 is wired for displaying the degree centigrade

symbol (°C). CD4511 is a BCD-to-7-segment display driver. IC2 and IC3 drive display DIS3 and DIS2, respectively.

Software program

The code for the microcontroller is written in 'C' language in Code Composer Studio version 5 (CCS v5) integrated development environment (IDE). CCS is a complete, Eclipse-based IDE that supports all the MSP430 microcontroller devices.

Code Composer Studio. When using CCS with an MSP430 MCU, a unique and powerful set of plug-ins and tools are made available to fully leverage the MSP430 microcontroller. Free and paid versions are available for download from Texas Instruments' website www.ti.com/tool/ccstudio-msp430. There are two options in the free version, each of which has some limitation:

1. **16kB code-limited version.** It has no time limit but supports firmware up to 16kB size only. All the MSP430 devices are supported.
2. **180-day time-limited version.** It has no code-size limit but is available only for 180 days (with registration).

All the MSP430 devices are supported.

The code for this project is 3kB only, so you can use any of the above free versions.

Operation of CCS. 1. Start Code Composer Studio (CCS) by double-clicking the icon on the desktop or selecting it from the Windows Start menu. When CCS loads, a dialogue box will prompt you for the location of a workspace folder. Browse to "C:\MSP430_LaunchPad\WorkSpace" and do not tick the "Use this as the default ..." checkbox. Click 'OK'. The workspace is saved automatically when CCS is closed.

2. Click 'New Project' option. A project contains all the files you will need to develop an executable output file (.out) which can be run on the MSP430 hardware. To create a new project, click "File→New→CCS Project." In 'Project Name' field, type 'Temperature_Indicator.' Uncheck "Use default location" box.

3. Next, select the appropriate device family, variant and connection type from the pull-down list. This will select the appropriate linker command